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TRANSFORMING THE POTHWAR PLATEAU: SLOPE AND WATER-CLASS FRAMEWORK FOR SOIL AND WATER CONSERVATION IN RAINFED PUNJAB, PAKISTAN

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ABSTRACT

The Pothwar Plateau, covering approximately 1.82 million hectares in northern Punjab, Pakistan, is one of South Asia's largest contiguous rainfed (barani) farming systems and receives no perennial river or canal supply across most of its cultivated area. Its undulating topography, erratic rainfall regime, fragile loess-derived soils, and centuries of subsistence cultivation have produced a degraded agro-ecological landscape in which about 20% of land is affected by accelerated water erosion and an estimated 4.2 billion cubic metres of water is lost annually as surface runoff. These stresses are amplified by documented climate change, manifesting as increased heat-wave frequency, flash floods, droughts and shifting monsoon patterns. Quantitative climatic record of Soil and Water Conservation Research Institute (SAWCRI), Chakwal along with a field-ready soil and water conservation strategy grounded in three decades of applied research showed that the area yields a mean annual rainfall of 655 mm with high inter-annual variability (coefficient of variation = 31%), a pronounced monsoon concentration (nearly 65% of annual rainfall in June-September), and a warming signal in the recent temperature series. Against this baseline, the review presents an evidence-based transformation plan centered on two cross-cutting field variables: (i) slope class, which governs erosion hazard and mechanical conservation requirements, and (ii) water-availability class, which governs cropping intensity and crop choice, together with a formal land-capability (soil-quality) classification linking four soil categories to specific crop, orchard, medicinal-plant, and grazing prescriptions. Detailed planning and guidance is presented for (i) mechanical conservation involving bunds, terraces, and safe water-disposal structures; (ii) land productivity management involving slope-differentiated soil-improvement techniques including microbial (PGPR) and biofertilizer inoculation; and (iii) an explicit agroforestry-exclusion and inclusion policy (i.e., eucalyptus-exclusion) for arable land. The plan is further extended by a documented assessment of climate-change impacts on Pothwar and a corresponding set of SAWCRI-recommended adaptation strategies, and by a value-added rural-industries component, apiculture, sericulture, lac culture, medicinal-plant processing, olive-oil production, biogas from livestock and crop waste, and agri-tourism, designed to diversify farm income on slopy, water-scarce land while reinforcing soil and water conservation objectives. The proposed implementation framework covers stepwise adoption, stakeholder roles, monitoring indicators, and climate-risk adaptation. Overall management recommendations are applied within the Murabba grid and calibrated to five slope classes (0-2%, 2-7%, 7-15%, and >15% eroded gullied land, and zigzag dissected land) with further four soil-capability categories, and four rainfall-availability zones, supported by international soil and water conservation literature.

Keywords: *Pothwar; Rainfed agriculture; Land-capability classification; Slope-based land-use zoning; Bund design; Bench terracing; Agroforestry exclusion; Drought adaptation.*

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INTRODUCTION

Geography, Boundaries, and Extent of Pothwar Punjab

The Pothwar Plateau is an undulating sub-Himalayan plain

situated at latitudes 32°10'-34°09' N and longitudes 71°10'-73°55' E in the northernmost Punjab Province of Pakistan, lying between the Indus and Jhelum rivers (Rashid et al.,

1997). Geologically part of the Indo-Gangetic uplands, which was elevated at the end of the Tertiary Period and is bounded by the River Jhelum to the east, the River Indus to the west, the Himalayan foothills to the north, and the Salt Range to the south (Encyclopaedia Britannica, 2024). Elevation ranges from about 300 to over 600 m above sea level, and the landscape is a dissected plateau of low hills, ravines, and undulating to steeply sloping terrain interspersed with relatively flat valley floors (Nizami et al., 2004). The plateau encompasses Rawalpindi, Attock, Chakwal and Jhelum districts, with adjoining barani tracts of Khushab, Mianwali, and the Islamabad Capital Territory (Ahmad, 1990; Shafiq et al., 2005). Total Pothwar rainfed tract comprises approximately 2.23 million hectares.

Land Cover Dynamics and Soil Classification of Pothwar Punjab

Modern geospatial technologies have transformed understanding of Pothwar's land-cover dynamics. Sentinel-2 imagery analyzed between 2017 and 2023 showed that 399 km² of fertile cropland was lost to urban encroachment within six years; weighted-overlay analysis classified 3719 km² as highly suitable for agriculture, 9809 km² as moderately suitable, 4069 km² as marginally suitable, and 4658 km² as unsuitable (Bhatti et al., 2023). RUSLE-GIS modelling detected high (30–100 t ha⁻¹ yr⁻¹) and very high (>100 t ha⁻¹ yr⁻¹) erosion-risk zones over 3% and 2% of the Rawal watershed, and over 29% and 14% of the Ghabbir watershed, respectively (Shafiq & Ahmad, 2017). Comparable remote-sensing-led dryland assessments have guided interventions in Jordan, Morocco, Ethiopia, and the Chinese Loess Plateau (Liu et al., 2006; Nyssen et al., 2004; Ouassar et al., 2009; Oweis et al., 2012).

Soils across Pothwar are predominantly loess-derived silt loams and silty clay loams and also sandy textures, generally low in organic matter and structurally fragile, making them highly susceptible to sheet and gully erosion once cover is removed or slopes are cultivated up-and-down rather than on the contour (Morgan, 2005). They fall under three primary USDA soil orders: Inceptisols (dominant), Entisols (eroded footslopes and ravine margins), and Aridisols (south-western semi-arid zone), with traces of Alfisols in sub-humid northern areas (FAO, 2006; Nizami et al., 2004). Soils are predominantly calcareous, with alkaline pH (7.8-8.5) that rises further with erosion (Miller et al., 1984; Shaheen et al., 2008). Major series include Missa (Typic Ustochrept), Rajar (Typic Ustorthent), and Guliana (Udic Haplustalf), the last being the most productive, found in sub-humid higher-rainfall zones (Rashid et al., 1997; Shaheen et al., 2008).

Agro-Climatic Characterization and Cropping Systems of the Pothwar

Pothwar Punjab unlike the canal-irrigated Indus plains; has no perennial supply reaching most of its cultivated area, and crop production is massively dependent on direct rainfall, supplemented locally by groundwater, small dams and ponds (Ashraf et al., 2000; Khan, 2002). Climatic records of Soil and Water Conservation Research Institute, Chakwal indicate that rainfall and temperature parameters govern crop selection, conservation-structure design, and climate-adaptation planning. Over the 49-year period 1977-2025, mean annual rainfall at Chakwal was 655 mm, with a median of 675 mm and a standard deviation of 201 mm. The coefficient of variation of approximately 31% confirms the high inter-annual unreliability that is the defining hydrological hazard of rainfed farming on the plateau. Annual totals ranged from a severe-drought minimum of 225 mm in 1979 to an exceptional maximum of 1315 mm in 2020 a nearly six-fold spread that dictates a conservation strategy built around both excess rainfall in monsoon and drought buffering in the later period (post monsoon). Regional analyses of the Soan River Basin confirm that rainfall across Pothwar is highly variable in space and time, with most highland stations showing decreasing annual trends and central and lowland stations increasing trends (Hussain et al., 2021). Climatic records indicate that although some years were too drier against wetter year: therefore Conservation planning must therefore treat every season as potentially dry.

The dominant Rabi crop is wheat, frequently rotated with gram (chickpea) and lentil on lighter, drought-prone land; the dominant Kharif crops are groundnut for which Attock and Chakwal are especially important together with sorghum, millets, mung bean, and maize in wetter northern tracts. Mustard/rapeseed is grown as an oilseed and fodder crop, and government-supported olive cultivation has expanded orchard area on marginal land. Livestock rearing on communal hill land and crop residues remains integral to the system and a critical buffer against crop failure.

Key Land and Water Productivity Constraints of Pothwar Punjab

Four structural constraints dominate in the Pothwar Punjab: (i) water scarcity, since rainfall meets only a fraction of full crop water requirement and most high-intensity summer rainfall is lost as runoff; (ii) erosion risk, driven by erodible loess, dissected terrain, and intense monsoon storms; (iii) declining soil health, from continuous cropping with minimal organic-matter return; and (iv) socio-economic limits, including rural out-migration reducing conservation

labour, fragmented smallholdings raising per-hectare structure costs, weak market access for high-value produce, and limited credit and risk-bearing capacity.

Soil Constrains - Soil Erosion, Nutrient Loss, and Soil Health

Soil erosion is the paramount biophysical crisis of Pothwar; sheet, rill, and gully erosion interact across the loess landscape to remove an estimated 20% of land from productive use (Ashraf et al., 2000). RUSLE modelling shows high to very high erosion risk dominating the medium- and low-rainfall zones where vegetation cover is sparse (Shafiq & Ahmad, 2017). Globally, land-use change has driven a measurable increase in soil erosion, with the most severe increases projected for Sub-Saharan Africa, South America, and Southeast Asia (Borrelli et al., 2017) analogous loess soils on the Chinese Loess Plateau exhibit comparably severe rates (Liu et al., 2006).

Surface soil loss is inseparable from nutrient depletion, because fine particles carry the highest concentrations of N, P, K, Zn, B, and organic matter. Centuries of cropping without adequate fertilization have produced universal macronutrient and micronutrient deficiencies across all Pothwar series (Rashid et al., 1997; Shaheen et al., 2008). Organic matter in eroded Missa and Rajar series falls below 0.5%, impairing biological activity, water-holding capacity, and aggregate stability (Lal, 2009). SAWCRI long-term fertility trials show that a combined organic-inorganic strategy consistently outperforms either alone, restoring productivity on degraded soils within 3-5 years.

The cumulative effect of these interacting stresses is a steady, self-reinforcing decline in overall soil health across Pothwar's cultivated landscape. As erosion strips nutrient-rich topsoil and organic matter falls below critical thresholds, soil structure degrades, infiltration capacity drops, and the land's resilience to drought and intensive cropping weakens further (Lal, 2009). This decline is not static but progressive: each cropping cycle without adequate

replenishment deepens the nutrient deficit and accelerates structural breakdown, pushing marginal soils of the medium- and low-rainfall zones closer to irreversible degradation (Shafiq & Ahmad, 2017).

Water Productivity Constraints of Pothwar Punjab

The hydrological signature of Pothwar is high peak runoff and low baseflow, a consequence of low organic matter, surface compaction, and limited vegetative interception on cultivated slopes (Ashraf et al., 2000). Rainfall-simulation experiments on representative soils show a bare Missa soil generating runoff within 8-12 minutes of 60 mm h⁻¹ rainfall, whereas the same soil under full grass cover delays runoff onset by 35–50 minutes. Studies across the Mediterranean, the Sahel, and dryland India confirm that vegetation cover is the single most powerful control on runoff generation (Casenave & Valentin, 1992; Cerdà et al., 2012; Pathak et al., 2005). Zone-specific rainfall, runoff coefficient, and effective rainfall are given in Table 1.

This runoff-dominated hydrology translates directly into moisture stress at the crop root zone, since water that leaves the field as surface flow within minutes of rainfall onset never becomes available for infiltration and storage in the profile. Even where seasonal rainfall totals appear adequate, the concentration of Pothwar's rainfall into high-intensity, short-duration events mean that a large share of the annual total is lost before crops can draw on it, leaving intra-seasonal dry spells that coincide with critical growth stages such as tillering, flowering, and grain-fill. The resulting soil moisture deficits are compounded by low organic matter and poor structural stability, which reduce the soil's water-holding capacity even when infiltration does occur. Consequently, moisture stress in Pothwar is less a function of rainfall scarcity per se than of poor rainfall-to-soil-moisture conversion efficiency, underscoring why interventions that raise runoff onset time and infiltration opportunity (vegetative cover, residue retention, contour treatments) are central to any water productivity strategy for the region.

Table 1: Zone-specific annual rainfall, runoff coefficient, and effective rainfall for the Pothwar Plateau.

Zone	Annual Rainfall (mm)	Runoff Coefficient	Effective Rainfall (mm)
A	850	0.25–0.35	550–640
B	600	0.35–0.45	330–390
C	400	0.45–0.55	180–220
D	200	0.55–0.65	70–90

Socio-Economic Limits

Beyond biophysical constraints, Pothwar's conservation and productivity gains are increasingly bounded by socio-economic limits. Sustained rural out-migration, driven by

limited on-farm returns, has depleted the labour pool traditionally available for terracing, bunding, and other conservation-intensive practices, leaving many structures unmaintained. Fragmentation of already-small holdings

across successive generations has driven up the per-hectare cost of soil and water conservation works, making individually-owned interventions economically unviable for a large share of farmers. Weak market linkages further discourage diversification into high-value horticultural and medicinal crops.

Field-Ready Planning and Guidance Framework Strategy for Pothwar Punjab

Redefining Land-Use Zoning Protocol and Murabba-Grid Field Planning

The foundational precursor to any effective resource management strategy in the Pothwar region is the precise, data-driven division of agricultural territory. Every downstream recommendation regarding bund alignment, terrace design, moisture conservation, safe water disposal, and crop choice depends on land first being surveyed, mapped, and zoned on a rigorous spatial grid rather than applied uniformly across a holding. We recommend that this grid be the Murabba: the traditional square survey block of 25 acres, subdivided into 25 one-acre units. Introduced during the British-era canal-colony settlement to rectangulate and administer land, the Murabba grid underpins the revenue maps of canal-irrigated Punjab but was never extended to most of the non-canal, rainfed Pothwar tracts, whose revenue records remain based on irregular, historically-accreted field boundaries. This absence of a regular cadastral grid is a specific, correctable weakness of the Pothwar revenue system: without it, conservation planning, runoff-management responsibility, and intervention accounting have no consistent spatial reference. Establishing a Murabba grid across the plateau with the killa (one acre) as the working planning unit within each square is therefore proposed as the foundational protocol of the transformation strategy, consistent with international guidance that soil-conservation planning units should match the scale at which slope, soil, and drainage actually vary (FAO, 2016; Morgan, 2005).

Slope-Based Land-Use Zoning of Pothwar Region

Slope is the primary zoning variable because it directly determines both the runoff and erosion hazard of a parcel and the practicality of different conservation structures (Morgan, 2005). The classification below moves from

flatter, intensively croppable land toward steeper land that should be progressively shifted away from annual tillage toward perennial vegetation and grazing (Table 3). It integrates the five-class SAWCRI slope framework with the operational thresholds used by the Soil Conservation Department, and is applied at the killa (acre) level within the Murabba grid per the protocol recommended.

Slope < 2%: High-Value Crops and Vegetables

Land in this class has the lowest erosion hazard and the best trafficability and water-retention characteristics, making it the plateau's most valuable land for intensive, high-value production wherever water availability permits. Recommended land uses include vegetables (onion, garlic, tomato, potato, pea, chilli, okra, cucumber, radish, carrot, spinach); high-value fruit such as nectarine, peach, dragon fruit, avocado, blueberry, guava gardens and watermelon on the flattest, best-drained pockets; and field crops (wheat, groundnut, gram, lentil) as rotation partners. Coriander, fenugreek (methi), mint, parsley, basil, green chilli, spring onion, well suited to small plots on Class I land adjacent to homesteads, with supplemental irrigation from farm ponds. Class I land should be prioritized exclusively for high-value vegetable and garden production rather than staple field crops, which can be grown on adjacent Class II slopes. Vegetable production on Class I land can be directly linked to weekly rural markets (mandis) in Chakwal, Rawalpindi, and Attock districts. SAWCRI recommends farmer group formation for collective marketing, cold storage, and value-added processing (sun-dried tomatoes, pickled vegetables, dried herbs) from Class I kitchen and market garden produce. On this class, contour (level) bunds are generally sufficient because natural slope is minimal and the priority is maximizing in-situ infiltration. Mulching, reduced tillage, soil-test-based split fertilization, and drip or bed-furrow irrigation conserve moisture on these high-value plots.

Precision laser levelling on Class I eliminates micro-topographic irregularities, achieves water savings of 25–30%, and increases wheat yield by 10–20% (Ali et al., 2022). SAWCRI recommends firm field bunds 30-45 cm height with compacted side slopes seeded with Mott grass or Super Napier, with grassed waterways or stone spillways at all bund outlets.

Table 2: Recommended boundary, border-strip, and kitchen-plot species for Class <2% land (intensive cropping; species confined to field edges, bunds, and homestead plots so the arable core stays in high-value crops).

Category	Species (local -botanical)	Role on <2% land	Placement / notes
Border grass	Khabbal (<i>Cynodon dactylon</i>)	Binds field bunds and channel edges; grazing aftermath	Bund faces and grassed waterways; not in the cropped

Border grass	Dhaman / buffel (<i>Cenchrus ciliaris</i>)	Drought-tolerant bund and border cover; fodder	bed (Malik et al., 2015)
Boundary tree	Shisham (<i>Dalbergia sissoo</i>)	Boundary shade, timber, windbreak; deep-rooted, non-competing at edges	Field margins on drier Class <2% pockets (khan et al., 2025)
Homestead medicinal	Aloe vera (<i>Aloe barbadensis</i>)	High-value gel crop for kitchen plots and marginal corners	Field boundaries and watercourse banks only (Lal, 2009)
Homestead aromatic	Mint (<i>Mentha longifolia</i>)	Culinary/medicinal; moist border and kitchen-garden strip	Homestead and border beds; drip-irrigated Damp field edges and homestead plots (Marwat et al., 2011)

Slope 2–7%: Field Crops, Medicinal Plants, Olive, Fig, and Ber

The gently to moderately sloping undulating uplands (2–7%) constitute the core agricultural zone of Pothwar. Without soil and water conservation, erosion rates at these slopes can exceed 15-25 t/ha/year under Pothwar's monsoon rainfall intensity (Shafiq & Ahmad, 2017). SAWCRI's primary SWC package for this class: broad-base contour bunds at 20-30 m vertical intervals, deep ploughing, composting, recommended NPK fertilization, and double cropping where rainfall permits. Wheat (Rabi, varieties Chakwal-50, Fatehjang, AARI-2011), Maize (Kharif, CIMMYT drought-tolerant hybrids), and Groundnut (Kharif, varieties GOLDEN, Peanut-1) are SAWCRI's principal field crop recommendations for Class II slopes. Sorghum (drought years, dual grain-fodder use), Pearl Millet (Zone C low-rainfall areas), Sunflower (emerging oil crop), Chickpea (Desi, 'Chakwal-86', Kabuli, excellent for Rabi fallow rotation on 2-5% slopes), and Lentil (Rabi rotation on lighter Missa soils). SAWCRI recommends wheat-groundnut and wheat-chickpea as the primary rotations for sustaining soil fertility on Class II slopes. The field crops like Mustard/Canola (Rabi, well adapted to calcareous Missa soils), Linseed (Rabi, dual oil-fibre crop), Mungbean (short-duration Kharif on 2–5% slopes after wheat harvest), and Soybean will diversify income while maintaining complete soil cover during the erosion-vulnerable Kharif season.

Berseem (*Trifolium alexandrinum*, Rabi fodder on well-

bunded Class II fields), Cowpea (Kharif green manure–fodder dual use), and Oats (*Avena sativa*, Rabi season) complete the field crop rotation cycle for Class II slopes. These crops protect the soil surface during the inter-season period and provide livestock fodder that is critical for the mixed farming systems of Pothwar.

SAWCRI's long-term fertility trials confirm that green manuring with Guar or Sesbania before wheat sowing increases field crop yield by 15–20% and reduces chemical N requirement by 30–40% on Missa series soils. Combined organic–inorganic fertilization and gypsum application (750 kg/ha every 2–3 years) are essential for sustaining Class II field crop productivity over the long term.

Recommended Soil and Water Conservation Technology

Broad-based, graded contour bunds at 20–30 m vertical intervals are the primary mechanical structure on 2–7% slopes, sized to safely convey peak monsoon runoff to a grassed waterway or stable outlet rather than allowing uncontrolled discharge (Morgan, 2005). Bund crests and risers should be compacted and immediately stabilized with vetiver or Napier/Mott grass, which extends functional bund life beyond 15 years and reduces total soil loss by 60–85% relative to unprotected earthen bunds (Truong et al., 2008; Shaheen et al., 2011). This should be paired with periodic deep/chisel tillage and farmyard manure or green-manure incorporation (guar, sesbania), which raises soil organic carbon and plant-available water-holding capacity on the fragile, loess-derived Missa soils that dominate this zone (Rawls et al., 2003; Shaheen et al., 2011).

Table 3: Recommended perennial, medicinal, aromatic, and bund-stabilizing species for Class 2–7% land (orchard and agroforestry zone; medicinal/aromatic plants grown as inter-row and understorey crops between olive, fig, and ber).

Category	Species (local — botanical)	Role on 2–7% land	Placement / notes
Orchard tree	Olive (<i>Olea europaea</i>); wild olive/kau (<i>Olea ferruginea</i>)	Flagship perennial for calcareous slopes; oil and canopy cover	Contour rows on bench/graded terraces (Oweis et al., 2012)
Orchard shrub	Ber/jujube (<i>Ziziphus mauritiana</i>); fig	Drought-hardy fruit; year-round root	Contour rows; crescent

	(<i>Ficus carica</i>)	cover	micro-catchments
Medicinal (inter-row)	Asgand / ashwagandha (<i>Withania somnifera</i>)	High-value root crop; tolerates dry, marginal slopes	Inter-row strips between orchard trees (Arshad et al., 2014; Malik et al., 2015)
Medicinal shrub	Bhekar (<i>Justicia adhatoda</i>)	Respiratory-medicine shrub; also a live hedge and bund binder	Terrace risers and orchard boundary hedge (Nizami et al., 2004; Punjab Forest, Wildlife & Fisheries Department, 2026)
Aromatic grass	Khavi (<i>Cymbopogon jwarancusa</i>)	Aromatic oil; deep roots stabilize terrace risers	Bund crests and riser faces; dual erosion/oil use (Liu et al., 2006; Punjab Forest, Wildlife & Fisheries Department, 2026)
Medicinal herb	Sanna / senna (<i>Senna alexandrina</i>)	Laxative-drug leaf crop for warm, well-drained slopes	Inter-row on drier 2–7% pockets
Bund grass	Vetiver (<i>Chrysopogon zizanioides</i>)	Living barrier on bund crests; 70–95% erosion reduction	Contour hedgerows across the slope (Truong et al., 2008)

Slope 7–15%: Grasses and Native Tree Species, Olive

Above roughly 7% slope, sustained annual cultivation is not advisable given erosion risk and the difficulty of mechanical bunding; land should be progressively converted to permanent grass and native tree cover functioning as an erosion buffer, fodder and fuelwood resource, and biodiversity corridor. Bench terracing is the appropriate measure where the site is retained in productive use (fruit or fodder trees); where the objective is stabilization and afforestation, contour trenching is a lower-cost alternative well suited to native tree establishment (Dorren & Rey, 2004; Nyssen et al., 2004). Native species suited to Pothwar hill land include phulai (*Acacia modesta*), kikar (*Acacia nilotica*), shisham (*Dalbergia sissoo*) on deeper sites, and wild olive on rockier ground, with tree spacing of 3–5 m along the contour. Structural and biological measures should be combined rather than used alone structures control the immediate hydraulics while vegetation provides long-term, low-maintenance stabilization and the most gullied sections should receive check dams before adjacent slopes are terraced (Stokes et al., 2014; Truong et al., 2008). Establishment requires protection from free grazing for the first 2–3 years. SAWCRI recommended natural Pothwar primary native tree including Phulahi (*Acacia modesta*), Kao (*Olea ferruginea*) and Sumbal (*Tamarix aphylla*). Olive thrives on calcareous, alkaline, well-drained soils with 300–600 mm rainfall, a perfect match for Pothwar's Missa and Pirsabak series on 7–15% slopes. Semi-circular micro-basin terraces (2 m diameter, 30 cm height bund, central planting pit with compost) are SAWCRI's

recommended planting system for olive on Class III slopes. The indigenous wild olive (*Olea ferruginea* / *Olea cuspidata*, locally known as 'Kao') found naturally on Class III–V slopes of northern Pothwar and the Salt Range provides a valuable local rootstock for grafting commercial olive varieties. SAWCRI recommends collecting propagation material from naturally occurring Kao trees, raising nursery seedlings, and using them as drought- and frost-tolerant rootstocks for commercial varieties such as Coratina, Leccino, and Arbequina, reducing establishment cost and improving long-term resilience. The inter-row spaces of young olive orchards (Years 1–5, before canopy closure) provide an ideal micro-environment for medicinal plant cultivation including Ajwain (*Trachyspermum ammi*), Isabgol/Psyllium (*Plantago ovata*), Moringa (*Moringa oleifera*), Lemongrass (*Cymbopogon citratus*) and Aloe Vera (*Aloe barbadensis*) on Class III slopes. SAWCRI recommends planting medicinal herbs between olive rows during the establishment phase, generating income from medicinal plant sales while the olive orchard matures.

Recommended Soil and Water Conservation Technology

Bench terracing is the preferred structural technology where 7–15% land is retained under fruit or fodder trees, cutting soil loss by 72–88% relative to untterraced slopes; contour trenching is the lower-cost alternative where the goal is pure stabilization and afforestation (Dorren & Rey, 2004; Nyssen et al., 2004). Gullied sections within this class should be treated with check dams before the adjacent slope is terraced, since structures alone control the immediate

hydraulics while grass and tree cover provide the long-term, self-maintaining stabilization (Stokes et al., 2014). For olive establishment specifically, semi-circular micro-basin

terraces (2 m diameter, 30 cm bund, compost-filled planting pit) concentrate the limited rainfall at the root zone during the vulnerable establishment years (Oweis et al., 2012).

Table 4: Recommended native trees, range grasses, and browse shrubs for Class 7–15% land (permanent grass–tree cover; erosion buffer, fuelwood, and fodder rather than tillage).

Category	Species (local — botanical)	Role on 7–15% land	Placement / notes
Native tree	Phulai (<i>Acacia modesta</i>)	Dominant hardy hill tree; fuelwood, honey, N-input, soil binding	Contour planting 3–5 m spacing (Lal, 2009; Nizami et al., 2004)
Native tree	Kau / wild olive (<i>Olea ferruginea</i>)	Evergreen slope-stabilizer on rocky ground; browse and shade	Rocky upper slopes; grafting stock for olive (Liu et al., 2006; Nizami et al., 2004)
Native tree	Kikar (<i>Acacia nilotica</i>); shisham (<i>Dalbergia sissoo</i>)	Fuelwood and timber on deeper colluvial pockets	Deeper soils and gully margins (Lal, 2009)
Range grass	Palwan (<i>Dichanthium annulatum</i>); <i>Chrysopogon serrulatus</i>	Perennial cover binding terrace benches; palatable fodder	Contour benches and inter-tree strips (Liu et al., 2006)
Range grass	Dhaman (<i>Cenchrus ciliaris</i>)	Drought-tolerant tussock; rapid cover after first rains	Reseeding bare 7–15% slopes (Liu et al., 2006)
Browse shrub	Sanatha (<i>Dodonaea viscosa</i>)	Fast-establishing erosion shrub; live check and fuel	Gully shoulders and degraded slope faces (Liu et al., 2006; Nizami et al., 2004)
Browse shrub	Garanda (<i>Carissa spinarum</i>)	Spiny protective browse; stabilizes stony slopes	Slope margins; natural grazing barrier (Lal, 2009)

Slope >15%: Grazing-Focused Pasture

The steepest land should be managed principally as permanent pasture under controlled grazing, with any structural measures aimed at protecting soil cover rather than enabling cultivation. These severely eroded gullied lands require emergency stabilization before any productive restoration is possible. SAWCRI's emergency intervention package: (i) gully plugging with loose stone at gully heads; (ii) check dams at 15–25 m spacing along the gully floor; (iii) vetiver grass slips at 15 cm spacing on gully walls within 30 days of structural work; (iv) contour trenching on adjacent slopes. SAWCRI data confirm complete gully stabilization within 2–3 monsoon cycles when this full package is applied.

Grazing paddocks and access tracks should follow the contour never running straight up-slope, which rapidly initiates gully erosion and low, dense contour vegetative strips substitute for earthen bunds over large hill areas. Seasonal grazing plans should rest each paddock during the monsoon growth period so cover is densest during the highest-erosivity rainfall, with stocking rates set conservatively against dry-season carrying capacity. Any residual cultivated patches above 15% should be retired to permanent cover, bare patches reseeded with drought-tolerant grasses at the start of the rains, and existing gullies

treated with small check structures before grazing resumes. Dhaman (*Cenchrus ciliaris*), Khavi (*Cymbopogon jwarancusa*), Mallah / jharber (*Ziziphus nummularia*), Bhekar (*Justicia adhatoda*), Buffel Grass (*Cenchrus ciliaris*), Phulahi (*Acacia modesta*) and Kikar (*Acacia nilotica*) are recommended for gully floors and stabilized alluvial fans behind check dams.

Recommended Soil and Water Conservation Technology

On slopes exceeding 15%, the priority technology is gully stabilization rather than cultivation-oriented earthworks: loose-stone gully-head plugging, check dams spaced at 15–25 m along the gully floor, and vetiver grass slips set at 15 cm spacing on gully walls within 30 days of structural work together achieve full stabilization within 2–3 monsoon cycles (Truong et al., 2008; Stokes et al., 2014). Contour-aligned grazing tracks and dense vegetative strips substitute for earthen bunds across these large hill areas, while resting each paddock during the monsoon growth period protects regenerating cover during the period of highest rainfall erosivity (Liu et al., 2006).

Class V, Dissected Land (>15% Slope): Protective Conservation Zone

Highly dissected, zigzag ravine land with slopes exceeding 15–35% and negligible topsoil has no direct agricultural future. These areas must be legally protected from

cultivation, grazing, and fuel wood extraction. SAWCRI's recommendation: protective afforestation using drought-

tolerant pioneer species combined with complete grazing exclusion for 5–10 years.

Table 5: Recommended pasture grasses and browse/fodder shrubs for Class >15% land (controlled grazing and protective cover only; no cultivation).

Category	Species (local — botanical)	Role on >15% land	Placement / notes
Pasture grass	Dhaman (<i>Cenchrus ciliaris</i>); <i>Chrysopogon serrulatus</i>	Backbone perennial pasture; persists under low rainfall	Reseed paddocks; rest during monsoon growth (Liu et al., 2006)
Pasture grass	Khavi (<i>Cymbopogon jwarancusa</i>)	Unpalatable-when-mature aromatic; holds soil on steepest faces	Steep gully heads and ridge tops (Liu et al., 2006)
Fodder shrub	Mallah / jharber (<i>Ziziphus nummularia</i>)	Drought-hardy browse; fruit and dry-season fodder	Scattered through grazing land; live fence (Liu et al., 2006; Ouessar et al., 2009)
Fodder shrub	Bhekar (<i>Justicia adhatoda</i>)	Non-palatable protective shrub; slope cover and bee forage	Ungrazed protective strips and gully walls (Nizami et al., 2004)
Protective tree	Phulai (<i>Acacia modesta</i>); kau (<i>Olea ferruginea</i>)	Scattered canopy; shade, browse, long-term stabilization	Ridge and upper-slope afforestation, grazing-excluded 2–3 yr (Lal, 2009; Liu et al., 2006)
Range legume	Senji (<i>Melilotus</i> spp.); <i>Lespedeza</i> spp.	N-fixing understorey; improves sward quality	Oversown into rested paddocks (Liu et al., 2006)

Recommended Soil and Water Conservation Technology: On Class V zigzag/dissected land, where topsoil is essentially absent, the appropriate technology is protective afforestation with drought-tolerant pioneer species combined with complete grazing exclusion for 5–10 years, allowing natural vegetative recovery before any productive use is reconsidered (Nyssen et al., 2004; Stokes et al., 2014). Loose-stone check structures at gully heads and contour trenching on the least-dissected margins should be used in parallel to arrest further headward erosion while the exclusion period runs its course (Morgan, 2005; Pandey et al., 2008).

Land-Capability (Soil-Quality) Classification and Crop Matching

A second and complementary classification based on soil quality confirmed by laboratory soil testing determines which crop, orchard, or vegetation type each parcel should carry within its slope-appropriate structural design. This land-capability approach follows the internationally established principle, first codified in the USDA land-capability classification system, that land should be used according to its capability and treated according to its needs (Klingebiel & Montgomery, 1961). Four soil-capability categories are defined for Pothwar, moving from the most productive land to the most degraded, each assigned to farm soil-fertility-laboratory testing before allocation (Table 5). Classification into the four categories below should be

confirmed, site by site, through testing at a soil fertility laboratory present in each district of the Pothwar assessing texture, depth, organic-carbon content, macro and micronutrient status, and salinity/sodicity where relevant consistent with the documented finding that Pothwar soils show universal but spatially variable macro and micronutrient deficiency (Rashid et al., 1997; Shaheen et al., 2008).

Land-Capability Categories

This classification operates jointly with, not in place of, the slope-based zoning of Section 5. Land-capability class determines the crop or vegetation that fills that structural template. In practice, Category i and ii land is concentrated on Slope Class <2% and 2–7% parcels; Category iii land frequently coincides with Slope Class 2–7% and 7–15% parcels of lower soil quality; and Category iv culturable waste is concentrated on Slope Class 7–15% and >15% land.

Water-Availability-Driven Cropping Strategies

Within the slope classes that remain under annual cultivation (broadly <2% and, for hardier crops, parts of the 2–7% class), and within land-capability Categories i and ii, cropping intensity and crop choice should be further differentiated by water-availability class a practical classification combining effective rainfall, soil moisture-holding capacity, and any supplemental source available to the parcel.

Table 6: Land-capability (soil-quality) classification for the Pothwar Plateau and corresponding crop and vegetation recommendations.

Category	Soil / Land Description	Recommended Crops and Land Use
i. Good fertile soils	Deep, fertile, well-structured, well-drained soils confirmed by soil test as high capability (broadly comparable to LCC Classes I–II (Klingebiel & Montgomery, 1961))	High-value crops: vegetables, tunnel/protected farming, orchards
ii. Second category	Moderately fertile soils, average depth and water-holding capacity, moderate water availability	Single- or dual-cropping: wheat, groundnut, or mung bean; introduction and adjustment of soybean where soil and moisture conditions allow
iii. Weak soils, limited water	Shallow or lower-fertility soils, limited or no supplemental water	Medicinal plants; olive; falsa (<i>Grewia asiatica</i>); fig; pomegranate; toot/mulberry (<i>Morus</i> spp.); ber (<i>Ziziphus mauritiana</i>)
iv. Culturable waste	Land too poor, degraded, or steep for regular cropping but not entirely barren	Grazing grasses and grazing shrubs (Australian kikar, <i>Acacia saligna/salicina</i>); fodder shrubs such as Moringa (<i>Moringa oleifera</i>) and phulahi (<i>Acacia modesta</i>)

Class-I: Zone A, >750 mm

The northern and northwestern areas of Pothwar, principally Murree Hills, Kahuta, Gujar Khan tehsils, and sub-montane areas of Rawalpindi, represent the most productive agricultural zone. These humid sub-zones support deep, well-structured Alfisols (Guliana series) capable of double cropping and high-value horticulture.

Fields with a reliable supplemental source or consistently moisture-retentive profile can support two crops per year, but intensification should be conditional on verified soil health rather than assumed. Soil organic carbon and macronutrient status should be confirmed by soil test before double cropping, with declining fields first receiving organic amendment; adequate drainage must be confirmed to avoid transient waterlogging and secondary salinity. Candidate sequences include vegetable–vegetable or vegetable–field-crop rotations, establishment of olive, fig, citrus, high-value fruit on the best-drained pockets, and lower-risk wheat–groundnut or wheat–mung-bean options.

Vegetables

Tomato (Rabi–Kharif), Capsicum, Brinjal, Peas, Cauliflower, Cabbage, Spinach, Fenugreek, Potato, Strawberry (cooler sub-zones), Cucumber, French Beans, Okra, Turnip, Radish, Carrot, Beet

Fruit Orchards

Peach, Plum, Cherry (higher sub-zones above 800 m elevation), Apple (Murree foothills >900 m), Pear, Mulberry, Guava, Apricot, Loquat, Strawberry

Field Crops

Wheat (double cropped with Maize or Fodder Sorghum), Tobacco (traditional cash crop), Maize, Soybean, Sunflower, Linseed

Drip or bed-furrow irrigation stretches a limited source

across a larger area, scheduled against growth stage and observed soil moisture. Even this Class I fields should retain a lower-water-demand fallback crop and a buffer storage reserve against supplemental-source failure in severe drought years.

Class-I: Zone B, 500-750 mm

The central belt of Pothwar, including much of Chakwal, central Rawalpindi, and eastern Attock, is where SAWCRI is headquartered and where the bulk of SAWCRI's long-term research has been conducted. This is the largest water-availability class across Pothwar's cultivated area, comprising fields dependent on direct rainfall with average soil-moisture storage and no reliable supplemental source. Recommended rotations include wheat–groundnut (the classic Rabi–Kharif rotation well supported by BARI Chakwal variety releases), wheat–pulses (mung bean, mash) on slightly drier land, gram or lentil in place of wheat where wheat performs poorly, and brassica as an oilseed or dual-purpose fodder crop. Soil testing every 2-3 years and post-monsoon visual erosion monitoring should guide management. Fertilizer rates should be based on soil tests and calibrated to rainfed rather than irrigated yield potential, with nitrogen timed to expected rainfall and farmyard manure prioritized ahead of the Rabi crop for both nutrients and moisture retention.

Primary Crops

Wheat (Rabi), Groundnut (Kharif), Chickpea, Pearl Millet

Vegetable Gardens

Pumpkin, Bottle Gourd, Bitter Gourd (Kharif); Tomato (early Kharif); Onion, Garlic, Spinach, Coriander (Rabi); Radish, Carrot, Turnip

Fruit Trees

Olive, (varieties: Coratina, Leccino, Arbequina),

Pomegranate (Kandhari, Bedana varieties), Ber (*Ziziphus mauritiana*, Umran, Gola), Fig (Brown Turkey), Almond (northern Zone B margins), Jamun (*Syzygium cumini*)

Class-III: Zone C, 250-500 mm

The southwestern and central-southern Pothwar receives only 250–500 mm rainfall, placing it firmly in the category of marginal agricultural land. SAWCRI's research focuses on converting this perceived 'wasteland' into a productive medicinal plant and agroforestry zone.

This class requires a risk-first approach built around minimizing the chance and severity of total crop failure. Sorghum and pearl millet serve as drought-tolerant Kharif staples; short-duration wheat, gram, and lentil varieties released for the Pothwar rainfed tract are selected to complete grain fill on stored moisture; and rainfed-adapted groundnut varieties from BARI Chakwal are preferred over higher-water-demand alternatives.

On-farm micro-catchment harvesting, community small dams and ponds sited by GIS-based slope, soil, and drainage criteria (Oweis et al., 2012; Reij et al., 2009), and roof-top harvesting capture and concentrate the limited rainfall. Crop residue retained as surface mulch reduces evaporative loss during establishment. Risk is further managed by keeping a higher proportion of fallow or low-input pasture, diversifying across crops with different drought-response strategies, and prioritizing this class for drought-resilience programme support, since the return on such support is highest where baseline risk is greatest.

Class-IV: Zone D < 250 mm

The driest southwestern margins of Pothwar, transitioning toward the Thal desert in Khushab district, receive <250 mm rainfall and are essentially unproductive under current management. SAWCRI's position is categorical: even this most barren land has productive potential if appropriate species and technologies are applied. No area of Pothwar, including its driest, most barren, and most severely eroded lands, should be left without a productive management prescription. Pearl Millet (Bajra), only crop viable in Zone D in years with >250 mm; must be planted immediately after monsoon onset.

SAWCRI's most critical recommendation for Zone D is to convert these barren lands into large-scale medicinal and aromatic plant estates. These species including Aloe vera, Caper (*Capparis spinosa*, Kabra), and Lana (*Haloxylon salicornicum*, 'Lana shrub') etc. require no irrigation after establishment and produce high-value products from what is currently economically inert land.

Trees and Shrubs

Jand (*Prosopis cineraria*): The most drought-tolerant nitrogen-fixing tree of Pothwar's arid southwestern zone. Survives on <200 mm rainfall once established.

Mesquite (*Prosopis juliflora*): Fast-growing, extremely drought-tolerant colonizer. Used for Zone D barren land rehabilitation across Pakistan's Thal desert, Rajasthan India, and Sahel Africa.

Cenchrus ciliaris (Buffel Grass): SAWCRI's primary grass for Zone D barren land stabilization. Establishes from broadcast seed in first monsoon rain. Provides immediate ground cover. Used successfully across Australia's arid zone, Rajasthan, and Sahel Africa.

Phulahi (*Acacia modesta*): Even in Zone D, Phulahi can establish in micro-catchment niches where moisture accumulates. SAWCRI recommends planting in Zai pits with compost and hydrogel on Class I–II micro-areas within Zone D landscapes.

Rainwater Harvesting

Underground reservoirs (tankas): Traditional cemented underground water storage vessels of 5,000–20,000 litre capacity collecting rooftop and road runoff.

Road-side catchments: Zone D roads concentrate runoff. SAWCRI recommends off-road ponding basins at 500 m intervals along Zone D road margins, collecting road runoff for tree and shrub establishment.

Zai pits with hydrogel: SAWCRI's most innovative technology for Zone D establishment: 30 × 30 × 30 cm planting pits filled with compost and SAP hydrogel, used for establishing Jand, Phulahi, Moringa, Aloe vera, and Harmal on Zone D barren land. A single monsoon rain is sufficient for establishment if hydrogel retains moisture for root development.

Building on the five slope classes, four water-availability classes, and four land capability classes examined in detail in Sections 4, 5, and 6, an integrated land-use and conservation strategy must address their combined effect rather than treating each factor in isolation. Since slope, water availability, and land capability interact to determine a site's true production potential and vulnerability to degradation, appropriate land use and conservation measures can only be selected once these three dimensions are considered jointly. The following link <https://waqasnaseem313.github.io/PLUA/> will guide site-specific recommendations for sustainable land management across Pothwar combining these three factors.

A

Pothwar Land Use Advisor
SLOPE • SOIL-CAPABILITY • WATER-ZONE FRAMEWORK

Match every Acre to its correct land use.

Select slope class, soil capability and rainfall/water-availability zone to receive a conservation-aware crop, orchard, medicinal-plant or grazing recommendation based on SAWCRI Chakwal's research.

Field suitability assessment
Select the three field-assessable variables for the target acre within its Murabba grid.

1. Slope class

Select slope class

Governs erosion hazard and which mechanical structure applies.

2. Soil-capability category

Select soil capability

Confirm on-site with district soil-fertility-laboratory testing.

3. Water-availability / rainfall zone

Select rainfall zone

Combines effective rainfall, moisture storage and any supplemental source.

Get land-use recommendation Reset selection

B

Recommended land-use plan
Gently to moderately sloping core agricultural zone of Pothwar • Average fertility, depth and water-holding capacity • Zone B — 500–750 mm (Much of Chakwal, central Rawalpindi, eastern Attock (SAWCRI headquarters))

Class II — 2–7% slope Category II — Moderately fertile soil Zone B — 500–750 mm

BEST RECOMMENDED LAND USE

Field-crop rotation with olive / fig / ber orchard and medicinal inter-row on the contour

Recommended for Zone B — 500–750 mm: Wheat-groundnut, wheat-pulses (mung/mash), gram or lentil, brassica, chickpea, pearl millet. Broad-based, graded contour bunds at 20–30 m vertical intervals, immediately stabilised with vetiver or Napier/Mott grass — functional bund life >15 years and 60–85% less soil loss than bare earth. Pair with deepchisel tillage and FYM or green manure (guar, sesbania).

ALTERNATIVES / SPECIES OPTIONS

Wheat-groundnut, wheat-pulses (mung/mash), gram or lentil, brassica, chickpea, pearl millet; Olive (Coratina, Leccino, Arbequina), pomegranate (Kandhari, Bedana), ber (Umran, Gola), fig (Brown Turkey), almond (northern margins), jamun

Select alternatives according to market demand, available labour, planting material and farmer investment capacity.

Conservation / SWC technology

Broad-based, graded contour bunds at 20–30 m vertical intervals, immediately stabilised with vetiver or Napier/Mott grass — functional bund life >15 years and 60–85% less soil loss than bare earth. Pair with deepchisel tillage and FYM or green manure (guar, sesbania).

Water strategy

Pakistan's largest rainfed cropping class. Soil-test every 2–3 years with post-monsoon erosion monitoring; apply FYM ahead of the Rabi crop for both nutrients and moisture retention.

Soil management

Use soil-test-guided fertilizer, FYM, residue retention, legumes and crop rotation. Suited to single- or dual-cropping (wheat, groundnut, mung bean) with soybean where conditions allow.

Important: Avoid up-and-down cultivation. Keep trees on bund crests, terrace risers and boundaries — never scattered within the tilled crop stand.

Print recommendation

Figure 1: The Pothwar Land Use Advisor decision-support interface. (A) Field suitability assessment panel, where the user selects slope class, soil-capability category, and water-availability/rainfall zone for the target parcel. (B) Resulting land-use recommendation panel, showing the best-recommended land use, alternative species options, and the corresponding conservation, water, and soil-management guidance.

Policy and Field Guideline

Laser Levelling and Field Preparation

Laser-guided precision land levelling on level to gently sloping Pothwar fields (Class I and Class II) delivers measurable benefits: water savings of 20–30%, yield increases of 10–20% in wheat, and improved germination uniformity (Ali et al., 2022). SAWCRI recommends laser levelling as the first investment for all Class I and flat Class II land Water Zones A and B before any crop intensification

program. In India's Haryana state, laser levelling increased wheat yield by 11–17% and reduced irrigation applications by one cycle in the dryland fringe areas (World Bank, 2020).

Engineered outlet requirement for all earthworks

No field should be bunded, terraced, or shaped without a corresponding, engineered path for surplus water. Wherever graded bunds, bench terraces, or contour trenches are used, provision must be made to convey the excess runoff they

collect to a stable outlet without causing new erosion.

Grassed waterways

Shallow, wide, vegetated channels along natural drainage lines, sized to carry the design peak flow at non-erosive velocity, should be preferred wherever terrain allows. They are low-cost and self-maintaining once grass cover is established, making them the default choice over engineered alternatives.

Lined or stone-pitched channels

These must be used where flow velocity or volume exceeds what a grassed waterway can withstand, typically at the outlet of larger graded-bund systems or bench-terrace drainage. They provide the structural strength needed where grass alone would erode.

Diversion drains

Construct the diversion drains above cultivated blocks. These diversion drains intercept runoff from adjoining steeper or uncultivated land before it enters the field system, preventing external runoff from overloading the field's own drainage.

Drop structures

Stone masonry structures dissipate flow energy wherever a waterway must descend a significant elevation change, preventing head-cutting that would otherwise progress upslope. Stilling basins or stone aprons at their base absorb the impact of falling water.

Check dams in incipient gullies

Within incipient gullies, check dams of brushwood, loose stone, or gabion (depending on catchment size) are spaced so that the crest of each is at or above the toe of the next one upstream, progressively raising the gully bed and halting downcutting (Morgan, 2005; Pandey et al., 2008).

Two-stage check dams

Two-stage check dams further improve sediment capture and recharge on gullied slopes, offering an enhanced version of the standard check-dam approach for more severe gully conditions (Hardie et al., 2012).

Grass strips and vetiver hedges on earthworks

Vegetative measures are the lowest-cost, most self-sustaining component of the erosion-control system and should always accompany, never entirely replace, structural bunds and terraces on actively farmed slopes. Grass strips along bund crests and terrace risers, established from deep-rooted drought-tolerant species, bind soil and reduce splash erosion on the earthwork itself; vetiver hedges are particularly effective, forming a dense living barrier that self-repairs and continues to function if the earthen structure is locally breached (Truong et al., 2008).

SAWCRI's field bund compacted with a mechanical

compactor and immediate seeding with Mott grass or Napier grass within 7 days of construction. SAWCRI data confirm that grass-covered bunds have a functional life of >15 years versus <5 years for bare earth bunds. The combination of firm bunds + grass cover + safe outlets reduced total soil loss by 60–85% in Pothwar experimental watersheds (Shaheen et al., 2011).

Hedgerow and alley planting

Hedgerow and alley planting of leguminous shrubs or fodder trees along contour lines provides an additional self-repairing barrier, complementing grass strips with a taller, woodier line of defense against erosion (Kang et al., 1990).

Permanent grass buffer strips and sediment-trapping vegetation

Permanent grass buffer strips at the downslope edge of every cultivated block intercept residual sediment-laden runoff. Sediment-trapping vegetation at the inlet to farm ponds extends their storage life by reducing the silt load entering the pond.

Direct disposed water into storage, not out of the watershed

Wherever the local catchment and terrain permit, water conveyed by grassed waterways, lined channels, or diversion drains should be directed into a farm pond, community pond, or small/mini-dam rather than allowed to leave the watershed as uncaptured discharge. This is consistent with the aquifer-recharge and storage benefits documented for Pothwar's small-dam network.

Requests routed through the Soil Conservation Department or Field Wing

Individual farmers are not expected to commission aerial or satellite survey themselves. Requests for new pond or dam sites should instead be routed through the Soil Conservation Department or Field Wing, which holds the GIS capacity and existing suitability layers, and can apply the six-step protocol above at no direct cost to the farmer.

Rationale for excluding trees from tilled cropland

Tree and shrub cultivation are recommended as the preferred land use for slope classes of 2–7% and 7–15%. However, on land currently under active, tilled (arable) cultivation, planting trees within the cropped area is not advised, because trees can cast shade that reduces sunlight available to crops, compete for soil moisture and nutrients through root spread, and obstruct machinery access during ploughing, planting, and harvesting. Collectively, these factors lower crop productivity and increase management complexity.

International agroforestry research consistently documents that scattered or in-field trees compete with adjacent arable

crops for light, soil moisture, and nutrients, with the balance of evidence showing a net yield penalty where trees are interspersed within, rather than confined to the margins of, actively tilled fields. Landscape-scale satellite and spatial-statistics analysis of parkland agroforestry found that tree canopy cover and density inside cultivated fields reduced crop yield through direct resource competition, even where trees at the field periphery had a positive buffering effect.

Studies of short-rotation alley cropping report reduced wheat, oilseed rape, and maize yield near tree strips, associated with greater soil-water tension at 30–60 cm depth from tree-root uptake and reduced solar radiation from shading.

Fertilizer-response studies in African parkland systems similarly report that dense in-field tree root systems can intensify below-ground competition for water and nutrients with the companion cereal crop, in some cases negating the benefit of applied fertilizer.

Trees on boundaries, not within the crop stand

Trees for fuelwood, fodder, timber, or fruit should be planted on field boundaries, bunds, or land already designated for perennial or forestry use, not scattered within an actively tilled arable field. Where boundary or windbreak trees are retained adjacent to arable land, a buffer zone should separate the tree row from the cropped area, and periodic root and canopy pruning should be used to reduce the competitive zone, consistent with agroforestry management guidance for reducing tree-crop competition. Land better suited to tree crops on evidentiary grounds (steeper slope, poorer soil) should be converted fully to orchard, fodder-tree, or forestry use, rather than interspersing trees into land retained for annual arable cropping.

Eucalyptus-Exclusion Policy on Arable Soils - Rationale and international evidence

A specific and stronger exclusion applies to Eucalyptus species, which should not be planted on arable soils in Pothwar that are under, or suitable for, crop production. Eucalyptus is documented internationally as capable of extracting water from shallow groundwater sources in arid and semi-arid environments, with plantation water use frequently exceeding local precipitation and measurable depression of the surrounding water table (Shi et al., 2012). Monitoring in Tehsil Swat Ranizai, District Malakand, found that the groundwater table declined more steeply in the immediate vicinity of Eucalyptus plantations than in comparable non-plantation areas over the 2000–2019 period, with locally severe declines recorded near plantation sites (Nasir et al., 2023). Approximately 80% of Pakistan's

land area lies within a semi-arid to arid climate classification, and Pothwar's own water balance is already tightly constrained by the $CV \approx 31\%$ rainfall variability documented in Section 1. The same groundwater-depletion mechanism therefore poses a direct threat to groundwater available for crop wells and small-dam recharge wherever Eucalyptus is planted near cultivated or water-storage areas. Eucalyptus should not be planted within, or immediately adjacent to, fields growing wheat, groundnut, vegetables, pulses, oilseeds, or any other arable crop, nor near wells, small dams, or ponds relied upon for irrigation or drinking water. For fuelwood, windbreak, or fodder needs, native and locally proven species phulai (*Acacia modesta*), kikar (*Acacia nilotica*), shisham (*Dalbergia sissoo*) on deeper sites, and Moringa (*Moringa oleifera*) should be used in place of Eucalyptus, consistent with the native-species recommendations of the area.

Crop Residue and Organic Matter Recycling

Three critical policy interventions recommended by SAWCRI:

Ban on use of cow dung as cooking fuel, replaced by subsidised LPG/biogas programs linked to mandatory compost pit construction on every farm

Ban on export and off-farm disposal of groundnut (peanut) straw from Chakwal district, requiring minimum 30% in-situ composting of groundnut straw annually

Mandatory crop residue recycling program: graduated incorporation of wheat and maize straw from 25% in Year 1 to 75% by Year 5, supported by subsidised straw choppers and compost inoculants developed at SAWCRI's laboratory

Climate Change: Impacts, Evidence, and Adaptation in Pothwar

Documented Climate Change Impacts

Pakistan ranks among the world's five most climate-vulnerable countries, having experienced the fifth-highest losses from extreme weather events between 1999 and 2018 despite contributing less than 1% of global greenhouse gas emissions. Punjab Province has demonstrated a temperature increase of approximately 0.97°C over the past century. For Pothwar specifically, documented climate change impacts include: increased heat-wave frequency and intensity; extreme monsoon flooding (the 2022 Pakistan mega-floods resulted in >1,700 fatalities and affected 33 million people, Otto et al., 2023); shifting rainfall distribution; increasing drought cycles in the southwestern zone; and paradoxical cold wave damage to early-flowering fruit trees in the sub-humid zone (Hussain et al., 2021).

Globally, comparable dryland and semi-arid regions are experiencing similar climate stress: the Sahel experienced

multi-decade rainfall decline of 25–30% from the 1950s to 1980s before partial recovery (Reij et al., 2009); Mediterranean dryland agriculture faces 20–40% yield reductions by 2050 under moderate warming scenarios (FAO, 2016); and South Asian dryland systems are projected to experience 10–25% increased rainfall variability by 2050 even where mean annual totals change little (Cairns et al., 2013).

Climate Change Adaptation Strategies, SAWCRI Recommendations

Heat-tolerant crop varieties

Heat tolerant varieties i.e. wheat varieties Chakwal-50 (heat-tolerant grain filling) and AARI-2011 should be preferred for the warming Pothwar growing environment.

Drought-tolerant horticultural species

Olive and pomegranate programs are inherently climate-change-adaptive, both species are native to Mediterranean climates with rainfall patterns similar to Pothwar's projected drier future.

Agro-meteorological early warning systems

It is recommended that a network of 30+ automatic weather stations across Pothwar, linked to SMS-based farmer advisories should be installed.

Agroforestry buffer zones

SAWCRI's recommendation for tree planting on Class III slopes (7-15%) creates landscape-level temperature moderation, reducing heat stress on adjacent Class II crop fields by 1.5–2.5°C on peak summer days.

Flood risk management

SAWCRI recommends upgrading all small dam spillways under climate change projections.

Medicinal plant diversification as climate resilience

SAWCRI recommends medicinal and aromatic plant enterprises as inherently climate-resilient income streams. Drought-adapted species (Harmal, Aloe vera, Capparis, Moringa) produce value even in below-average rainfall years, providing income continuity when cereal crops fail.

Drone-based slope reforestation

SAWCRI recommends procurement of agricultural UAV drones for broadcast seeding of grass (vetiver, buffel grass, *Cenchrus ciliaris*) and tree species (Phulahi, Kikar) on inaccessible Class IV–V gullied slopes where manual seeding is impractical. In China's Sichuan province, drone seeding reforested 33,000 ha of degraded terrain in a single season at one-sixth the cost of manual planting (Xinhua, 2020).

Community seed bank establishment

SAWCRI recommends a network of village-level community seed banks conserving traditional drought-tolerant Pothwar crop varieties. In Ethiopia's Tigray, 70

community seed banks conserving 500+ traditional drought-tolerant sorghum and barley varieties saved communities during extreme drought years when commercial varieties failed completely (FAO, 2014). SAWCRI recommends minimum one community seed bank per union council across Pothwar's six districts.

Agro-silvo-pastoral rotational grazing

SAWCRI recommends controlled silvopastoral rotational grazing on Class III slopes, dividing degraded slopes into paddocks with temporary electric fencing, rotating livestock every 15–20 days to prevent overgrazing and allow perennial grass recovery. In the semi-arid Deccan and Ethiopia's highlands, rotational grazing restored 30–50% perennial grass cover within two years on bare degraded slopes (Savory & Butterfield, 1999).

Value-Added Rural Industries

Apiculture (Honey Bee Farming)

Pothwar's diverse floral resources, from wild Brassica and mustard in winter to Acacia, Ziziphus, and Coriander in summer, create a rich multi-florals nectar calendar ideal for honey production. SAWCRI recommends promoting *Apis mellifera* (Italian honey bee) in Langstroth hives across the semi-arid zones, integrated with medicinal plant cultivation in Zone C. The combination of ajwain, coriander, Phulahi, and Ziziphus plantings recommended by SAWCRI for Zones B and C create a 10-month nectar flow supporting 10–20 hive colonies per ha of floral resources. Pakistan's honey production potential is estimated at 200,000+ tonnes annually against current production of ~8,000 tonnes (FAO, 2022). SAWCRI recommends training programs for Zone C farmers in hive management, honey processing, and wax production to capture the full value chain.

Sericulture (Silk Production)

Mulberry (*Morus alba*, *M. multicaulis*) grows vigorously on moist, well-drained soils in the sub-humid northern zone (Zone A and B). SAWCRI recommends expanding mulberry cultivation on terraced 7–15% slopes in Zone A, intercropped with shade-tolerant vegetables, as a year-round income enterprise combining erosion control and sericulture. PARC's sericulture program has demonstrated viable silk production in Murree and Gujar Khan tehsils. One kg of silk cocoons requires 12-15 kg of fresh mulberry leaves. SAWCRI's mulberry planting at 1,600 plants/ha on Class III Zone A terraced slopes can supply 15,000-20,000 kg fresh leaves/ha/year, sufficient for 1,200-1,500 kg fresh cocoons, with a gross return 3-4 times higher than equivalent cereal production.

Lac Culture

Pothwar's arid zone (Zone C and D) is home to extensive

natural populations of *Ziziphus* and *Acacia*, the ideal lac host species. SAWCRI recommends lac culture as a component of agroforestry systems on Class III and Class IV land in Zones C and D, requiring minimal capital investment and generating high returns per unit area in comparable dryland settings in Rajasthan and Bangladesh (Sharma et al., 2006).

Medicinal Plant Processing Enterprises

SAWCRI's medicinal plant program for Pothwar is not limited to farm-level production. The institute recommends establishment of: (i) village-level primary processing units (drying, cleaning, grading) for ajwain, kalonji, isabgol, chamomile, and harmal seed, reducing post-harvest losses from 25–40% to <10%; (ii) small-scale steam distillation units for essential oil extraction from ajwain (thymol), lemongrass, and peppermint, the 'bouillon cube' of rural income; (iii) Aloe vera gel extraction and stabilization units at the tehsil level, serving Zone C barren land reclamation gardens; (iv) Moringa leaf powder production for export nutrition markets.

Olive Oil and Table Olive Enterprise

SAWCRI's olive program for the Pothwar Plateau, anchored at Chakwal research station, represents one of the most significant agricultural transformation initiatives in Pakistan's dryland history. SAWCRI recommends zone B–C olive estates of 10–50 ha on Class II–III slopes with stone terraces and micro-irrigation from mini-dams. Center of Excellence for Olive Research (CEFORT)'s research has confirmed the production of high-quality oil with oleic acid content of 68–74%, comparable to Mediterranean standards, from Pothwar-grown olives.

Biogas from Livestock and Crop Waste: Slopy Farm Energy Independence

Biogas technology, addresses two critical problems simultaneously for Pothwar's slopy water-scarce farms: energy poverty and organic matter loss. Currently, 60–70% of cow dung on Pothwar farms is dried and burned as cooking fuel, destroying the organic matter and nutrients that are desperately needed to restore degraded slopy soils. A household biogas digester of 4–8 m³ capacity, fed with dung from 3–5 cattle, produces: 1.5–2.5 m³ biogas per day (sufficient for all household cooking); 15–20 kg nutrient-rich biogas slurry per day (far superior to raw dung as a soil amendment, with 40–60% higher plant-available nitrogen); and eliminates indoor air pollution from biomass combustion (Dhingra et al., 2014). The biogas slurry, applied at 5–10 t/ha/year on degraded Class II–III slope soils, restores organic matter 3–4 times faster than uncomposted dung because anaerobic digestion partially

humifies the organic matter, making it directly bioavailable. In China's hilly Loess Plateau, the national biogas program installed 42 million household digesters by 2012 in rural areas analogous to Pothwar in topography and farming system, simultaneously resolving cooking energy needs, restoring soil organic matter on slopy fields through slurry application, and reducing deforestation pressure for firewood (Zou et al., 2015).

Tourism Development in Pothwar

Archaeological and Historical Tourism

Taxila (Rawalpindi District): UNESCO World Heritage Site with Bhir Mound, Sirkap, and Sirsukh; Taxila Museum housing one of the finest Gandharan art collections in the world

Rohtas Fort (Jhelum District): 16th century Mughal fortification, UNESCO World Heritage Site

Ketas Raj Temple Complex (Chakwal District): Ancient Hindu pilgrimage site with sacred lake dating to pre-Islamic civilisations; of global spiritual and archaeological significance

Pharwala Fort (Rawalpindi): Pre-Mughal Hindu Sahi fort above the Haro River

Geological and Natural Landscape Tourism

Salt Range Geological Park: Natural geological museum of global educational significance; Khewra Salt Mine, world's second largest

Soan Valley Fossil Sites: Pleistocene pebble-tool sites representing South Asia's earliest known tool-making civilization

Shakarparian and Rawal Lake: Established urban green spaces with eco-tourism potential

SAWCRI Chakwal Farm: Agri-tourism potential as a model demonstration farm for olive orchards, medicinal plant gardens, soil conservation structures, and watershed rehabilitation, a living classroom for farmers, students, and policymakers

Agri-Tourism and Rural Tourism

Groundnut harvest festivals in Chakwal (October):

Experiential tourism around Pakistan's iconic Pothwar crop

Olive orchard tourism (Chakwal, Attock): Mediterranean-style landscape experiences around SAWCRI-promoted olive estates

Medicinal plant farm circuits in Zone C, linked to traditional hakeem (Unani medicine) demonstrations and SAWCRI's medicinal plant demonstration plots

SAWCRI Open Farm Days: SAWCRI Chakwal recommends quarterly open-farm demonstration events combining farmer training with agri-tourism, showcasing check dams, contour bunds, olive orchards, medicinal plant

gardens, and rainwater harvesting structures to domestic and international visitors

CONCLUSIONS

The Pothwar Plateau stands at a critical juncture. Its 1.82 million hectares face compound threats from accelerating soil erosion, nutrient depletion, groundwater decline, and climate change. The climatic record of the SAWCRI Chakwal station makes the challenge precise: a mean annual rainfall of 655 mm with 31% inter-annual variability, roughly two-thirds concentrated in a short and erosive monsoon, a pronounced post-monsoon deficit that constrains winter cropping, and a measurable warming tendency. These are the design parameters for every conservation structure and cropping decision on the plateau. Against this baseline, the transformation plan grounded in three decades of applied SAWCRI and BARI research and corroborated by international literature rests on a single organizing principle: no killa of Pothwar, however steep, eroded, arid, or barren, should be left without a productive and ecologically appropriate land-use prescription, correctly matched to its own measured slope and soil quality. By combining Murabba-grid field planning with three field-assessable variables slope class, land-capability class, and water availability together with validated engineering design, an explicit aerial/GIS siting protocol for water storage, an evidence-based exclusion of trees and Eucalyptus from arable land, and slope-differentiated soil-biological management including microbial inoculation, the strategy is directly actionable by the Soil Conservation Department, the Field Wing, and farmers themselves.

The plan also treats Pothwar's documented climate-change exposure, rising heat-wave frequency, flash floods, shifting monsoon timing, and deepening post-monsoon drought, as a standing design constraint rather than a future contingency, and pairs the conservation framework with SAWCRI's climate-adaptation recommendations. Recognizing that soil and water conservation alone cannot secure smallholder livelihoods on Pothwar's slopy, water-scarce land, the plan is completed by a value-added rural-industries strategy spanning apiculture, sericulture, lac culture, medicinal-plant processing, olive-oil production, biogas recovery from livestock and crop waste, and agri-tourism, each selected to generate income from the same land-capability and water-availability zones already prescribed for conservation, while returning organic matter, ground cover, or off-farm revenue to the system.

The technologies are proven, the species identified, the climatic baseline quantified, and the institutional roles

defined. The task ahead is coordinated implementation matching each killa of the plateau to its appropriate prescription, and each prescription to the climate the station record so clearly describes.

AUTHORS CONTRIBUTION

All authors contributed equally in the manuscript.

CONFLICT OF INTEREST

There is no conflict of interest among authors.

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